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4.6 The Rate and Extent of Chemical Change

AQA GCSE Chemistry (8462) — Paper 2

Chemical reactions happen at very different speeds — an explosion is over in a fraction of a second, while rusting takes months. This topic explains **how fast** reactions go (the **rate**), what we can do to speed them up or slow them down, and why, using **collision theory**. The second half looks at **how far** reactions go (the **extent**): many reactions are **reversible** and reach a state of **dynamic equilibrium**, and at Higher tier you must predict how changing the conditions shifts the position of that equilibrium. These ideas are essential for industry, where chemists must make products quickly, cheaply and safely.

Content marked **(HT)** is Higher Tier only. This whole topic is common to GCSE Chemistry and Combined Science; there is no triple-only content in 4.6 (the Haber process in detail is in 4.10).

4.6.1 Rate of Reaction

4.6.1.1 Calculating Rates of Reactions

> The rate of a chemical reaction can be found by measuring the **quantity of a reactant used** or the **quantity of product formed** over time:
>
> `mean rate of reaction = quantity of reactant used ÷ time taken`
>
> `mean rate of reaction = quantity of product formed ÷ time taken`

The quantity of reactant or product can be measured by the **mass in grams** or by a **volume in cm³**. The units of rate of reaction may be given as:

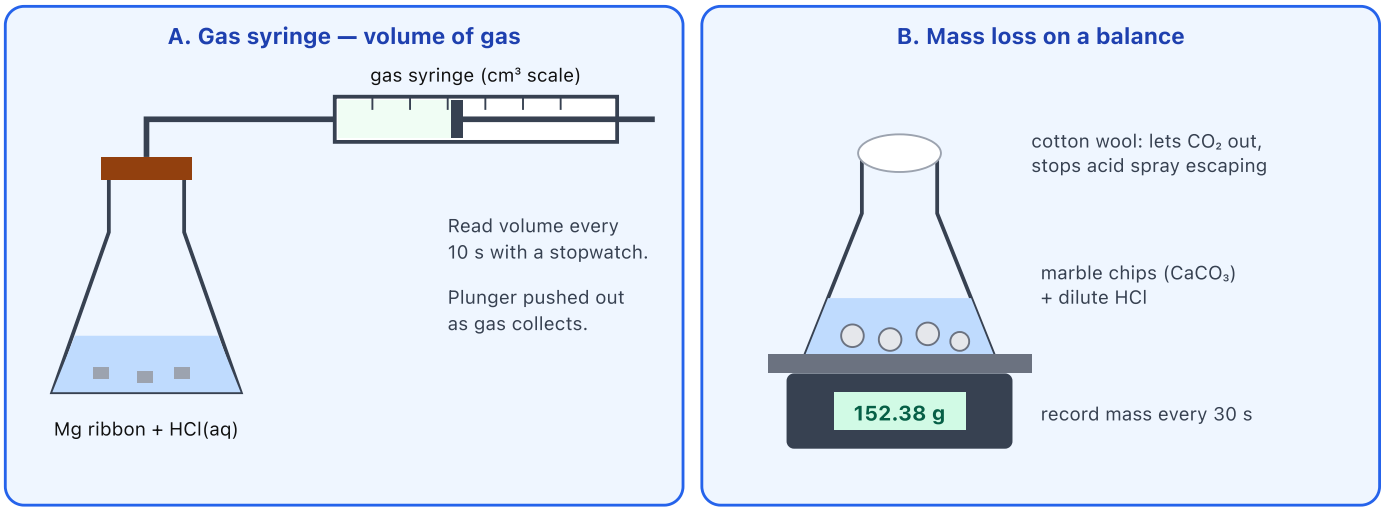
Quantity measured	Unit of quantity	Unit of rate
Mass of reactant lost / product formed	g	**g/s**
Volume of gas produced	cm^3	**cm^3/s**
(HT) Amount of substance	mol	**mol/s**

Measuring Rate in the Laboratory

Method	What is measured	Suitable for	Notes
Gas syringe (or upturned measuring cylinder/burette over water)	Volume of gas produced at regular time intervals	Reactions producing a gas, e.g. Mg + HCl, $CaCO_3$ + HCl	Very accurate; syringe is usually 100 cm^3 max. Upturned cylinder unsuitable for gases that dissolve in water (e.g. CO_2 is slightly soluble).
Mass loss on a balance	Mass lost as gas escapes from an open flask	Reactions giving off a dense gas, e.g. CO_2 from marble chips + acid	Use cotton wool in the neck to let gas out but stop acid spray. Hard to use with hydrogen (very light — small mass change).
Colour change / turbidity	Time for a solution to turn cloudy or change colour	e.g. sodium thiosulfate + HCl forming a sulfur precipitate	"Disappearing cross" method. Gives a single time; rate $\propto$ 1/time. Subjective end point — a light sensor improves this.

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Two ways of following a reaction that produces a gas

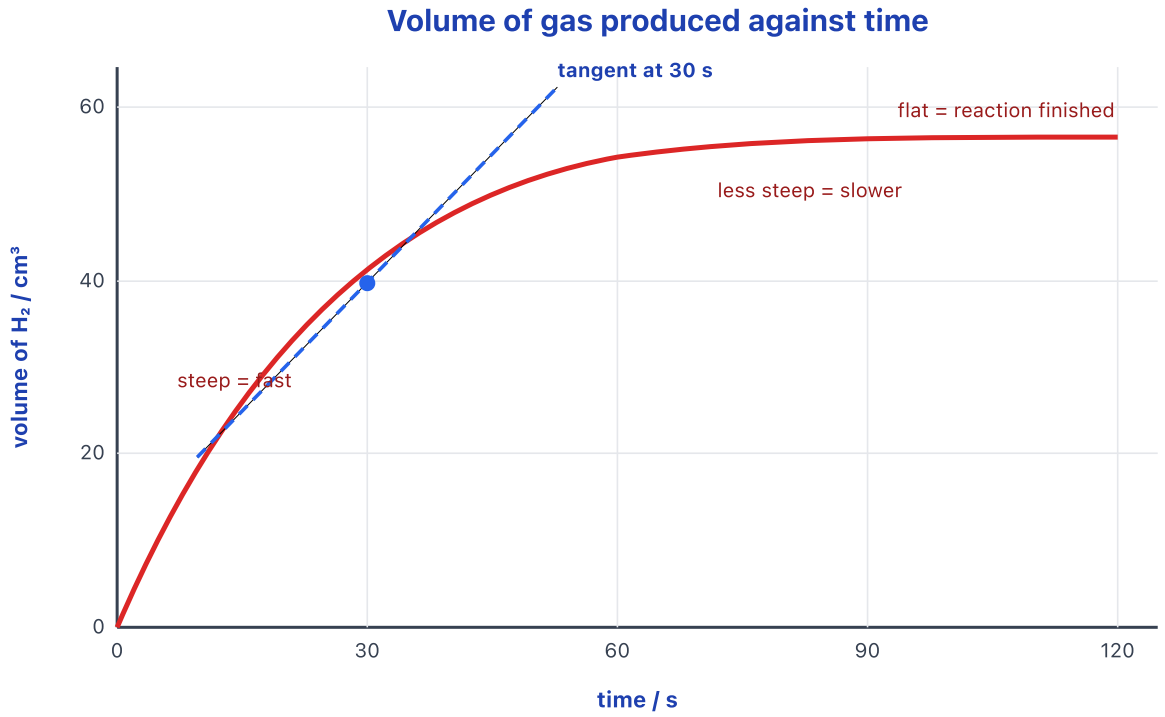


Graphs of Reaction Progress

Plotting the quantity of product formed (or reactant used) against time gives a curve. You should be able to draw and interpret graphs showing quantity against time, and use the **slope** (gradient) of the graph as a measure of the rate of reaction:

- The curve is **steepest at the start** — the rate is greatest because the concentration of reactants is highest.
- The curve gets **less steep** as reactants are used up — the rate decreases.
- The curve becomes **horizontal (flat)** when the reaction has **finished** — one reactant (the limiting reactant) has been used up completely.

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Worked Example 1 — Mean Rate

```
Question: Magnesium reacts with hydrochloric acid. 60 cm³ of
          hydrogen is collected in the first 30 s. A total of
          75 cm³ is collected after 120 s, when the reaction stops.
          (a) Calculate the mean rate over the first 30 s.
          (b) Calculate the mean rate over the whole reaction.

(a) mean rate = volume of product ÷ time
              = 60 ÷ 30
              = 2.0 cm³/s

(b) mean rate = 75 ÷ 120
              = 0.625 cm³/s ≈ 0.63 cm³/s

Note: the mean rate over the whole reaction is lower because
the reaction slows down as the reactants are used up.
```

Rate at a Specific Time — Tangents (HT)

Higher-tier students must be able to **draw tangents to the curves** on these graphs and use the **slope of the tangent** as a measure of the rate of reaction at a specific time.

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1. Place a ruler so that it just **touches the curve at the chosen time** without crossing it, with equal angles between the ruler and the curve on either side.
2. Draw a long straight line (a longer tangent gives a more accurate gradient).
3. Pick two points far apart on the tangent and read their coordinates.
4. gradient = **change in y ÷ change in x**. This is the rate at that time.

Worked Example 2 — Gradient of a Tangent (HT)

```
Question: A tangent is drawn to the volume–time curve at t = 30 s.
          It passes through (10 s, 20 cm³) and (53 s, 62 cm³).
          Calculate the rate of reaction at 30 s.

gradient = Δy ÷ Δx
         = (62 − 20) ÷ (53 − 10)
         = 42 ÷ 43
         = 0.977 cm³/s

Answer:  ≈ 0.98 cm³/s
```

Worked Example 3 — Rate in mol/s (HT)

```
Question: In a reaction, 0.48 g of magnesium (Ar = 24) is used up
          in 40 s. Calculate the mean rate in mol/s.

moles of Mg = mass ÷ Ar = 0.48 ÷ 24 = 0.020 mol
mean rate   = 0.020 ÷ 40 = 0.00050 mol/s  (5.0 × 10⁻⁴ mol/s)
```

Exam Tip: If asked to calculate a rate "at" a particular time, you **must** draw a tangent — dividing the y-value by the x-value at that point gives the *mean* rate from time zero, not the rate at that instant. Show your tangent clearly on the graph and show the triangle you used: the examiner will award marks for a correctly placed tangent even if your reading is slightly off. Always give units (cm³/s, g/s, mol/s).

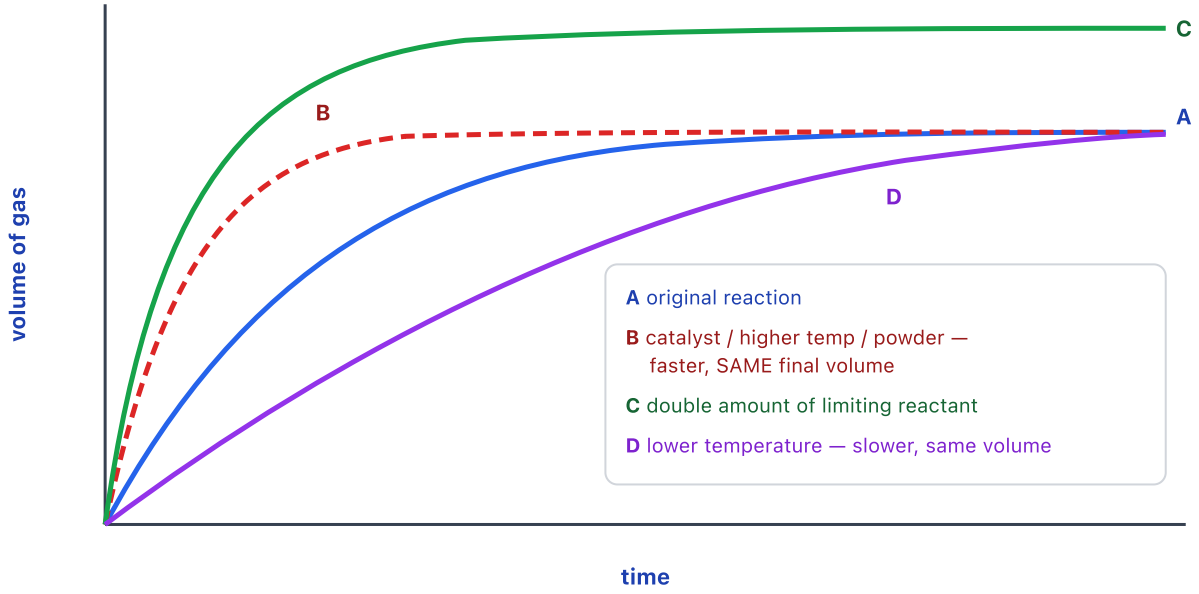
4.6.1.2 Factors Which Affect the Rates of Chemical Reactions

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Factors which affect the rates of chemical reactions include:

- the **concentrations** of reactants in solution;
- the **pressure** of reacting gases;
- the **surface area** of solid reactants;
- the **temperature**;
- the presence of **catalysts**.

Effect of changing conditions on the curve



Exam Tip: When sketching a curve for a faster reaction, make it **steeper at the start** and level off **sooner** — but at the **same height** if the amount of limiting reactant is unchanged. Only changing the *amount* of the limiting reactant changes the final volume of product. A catalyst, temperature or surface-area change alters how fast you get there, not how much you get.

Required Practical 5 — Rates of Reaction

Aim: to investigate how changes in concentration affect the rates of reactions by a method involving **measuring the volume of a gas produced** and a method involving a **change in colour or turbidity**. This should be an investigation involving developing a hypothesis.

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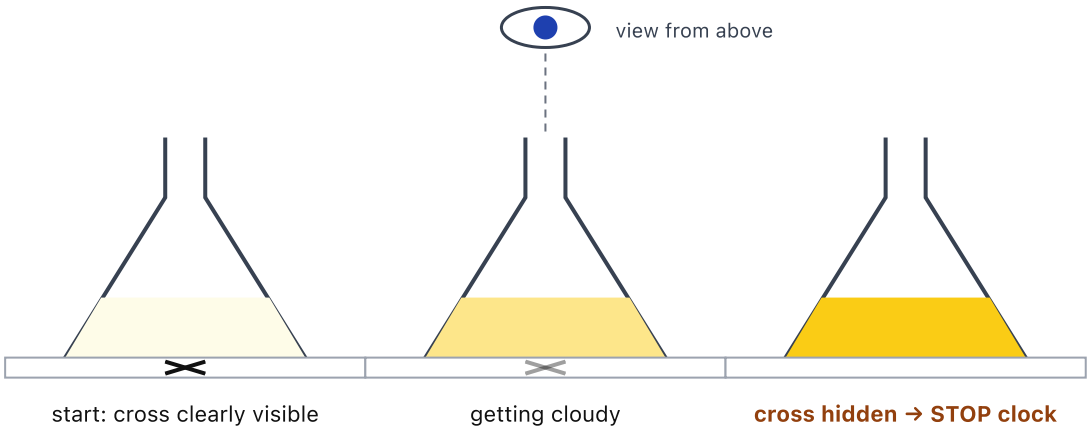
Method A — Turbidity: Sodium Thiosulfate and Hydrochloric Acid

$$Na_2S_2O_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + SO_2(g) + S(s) + H_2O(l)$$

↑ yellow sulfur precipitate makes the solution cloudy

1. Draw a black **cross** on a piece of white paper and place a conical flask on top.
2. Measure **10 cm³ of sodium thiosulfate solution** and **40 cm³ of water** into the flask (for the most dilute run), using measuring cylinders.
3. Add **10 cm³ of dilute hydrochloric acid**, swirl, and **start the stopwatch** immediately.
4. Look down through the flask from above. **Stop the clock** when the cross can **no longer be seen**.
5. Repeat with different concentrations of thiosulfate (e.g. 20 cm³ thiosulfate + 30 cm³ water, 30 + 20, 40 + 10, 50 + 0), keeping the **total volume the same**.
6. Repeat each concentration and calculate means. Calculate the rate as **1/time** (s^{-1}) and plot against concentration.

The "disappearing cross" experiment



Method B — Volume of Gas: Magnesium and Hydrochloric Acid

1. Set up a conical flask with a bung and delivery tube connected to a **gas syringe** (or an upturned measuring cylinder filled with water in a trough).
2. Measure **50 cm³ of hydrochloric acid** of a known concentration into the flask.

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3. Add a measured **3 cm length of magnesium ribbon**, immediately replace the bung and start the stopwatch.
4. Record the **volume of gas** every 10 seconds until no more gas is produced.
5. Repeat with different concentrations of acid. Plot volume against time for each, and compare initial gradients.

Variable	Method A (thiosulfate)	Method B (magnesium)
Independent	Concentration of sodium thiosulfate	Concentration of hydrochloric acid
Dependent	Time for cross to disappear	Volume of hydrogen produced (over time)
Control	Total volume; volume and concentration of HCl; temperature; same cross and flask; same observer	Volume of acid; length/mass of Mg; temperature; surface area of Mg

Safety: sulfur dioxide is produced in Method A, which is toxic and can trigger asthma — use a well-ventilated room and pour the mixture into a sodium carbonate solution afterwards. Hydrochloric acid is an irritant — wear eye protection. Hydrogen is flammable — keep away from flames.

> **Exam Tip:** In the thiosulfate experiment, the end point is **subjective** — different people judge "can't see the cross" differently. Improvements: use the same person each time, use the same cross, or replace the eye with a **light sensor and data logger**. Remember that **rate = 1 ÷ time**, so a *shorter* time means a *faster* rate.

4.6.1.3 Collision Theory and Activation Energy

> **Collision theory** explains how various factors affect rates of reactions. According to this theory, chemical reactions can occur **only when reacting particles collide with each other** and **with sufficient energy**.

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The minimum amount of energy that particles must have to react is called the **activation energy**.

Increasing the **concentration** of reactants in solution, the **pressure** of reacting gases, and the **surface area** of solid reactants increases the **frequency of collisions** and so increases the rate of reaction.

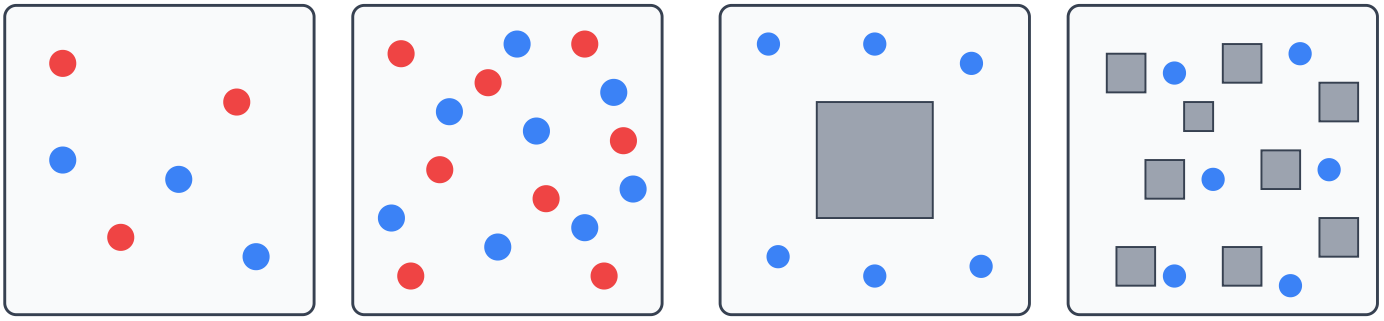
Increasing the **temperature** increases the **frequency of collisions** *and* makes the collisions **more energetic**, and so increases the rate of reaction.

A collision that has at least the activation energy and leads to a reaction is a **successful collision**. Most collisions are *not* successful — the particles simply bounce apart.

Factor increased	Effect on particles	Effect on rate
Concentration (solutions)	More reacting particles in the same volume → particles closer together → **more frequent collisions**	Increases
Pressure (gases)	Same number of gas particles in a smaller volume → **more frequent collisions**	Increases
Surface area (solids)	Breaking a solid into smaller pieces exposes more particles on the surface → **more frequent collisions** with the other reactant	Increases
Temperature	Particles move **faster** → **more frequent collisions**; **AND** more particles have energy ≥ activation energy → a **greater proportion of collisions are successful**	Increases (significantly)
Catalyst	Provides an alternative pathway with a **lower activation energy** → more collisions are successful	Increases

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Concentration and surface area — particle view



low concentration
few collisions

high concentration
more frequent collisions

one large lump
inner particles shielded

same mass as powder
larger surface area exposed

Surface Area to Volume Ratio

Cutting a solid into smaller pieces **increases its surface area to volume ratio**. You may be asked to calculate this:

```
Cube of side 2 cm:
  surface area = 6 × (2 × 2) = 24 cm²
  volume       = 2 × 2 × 2   = 8 cm³
  SA : V       = 24 : 8 = 3 : 1

Same cube cut into eight 1 cm cubes:
  surface area = 8 × 6 × (1 × 1) = 48 cm²
  volume       = 8 × 1           = 8 cm³
  SA : V       = 48 : 8 = 6 : 1   → doubled

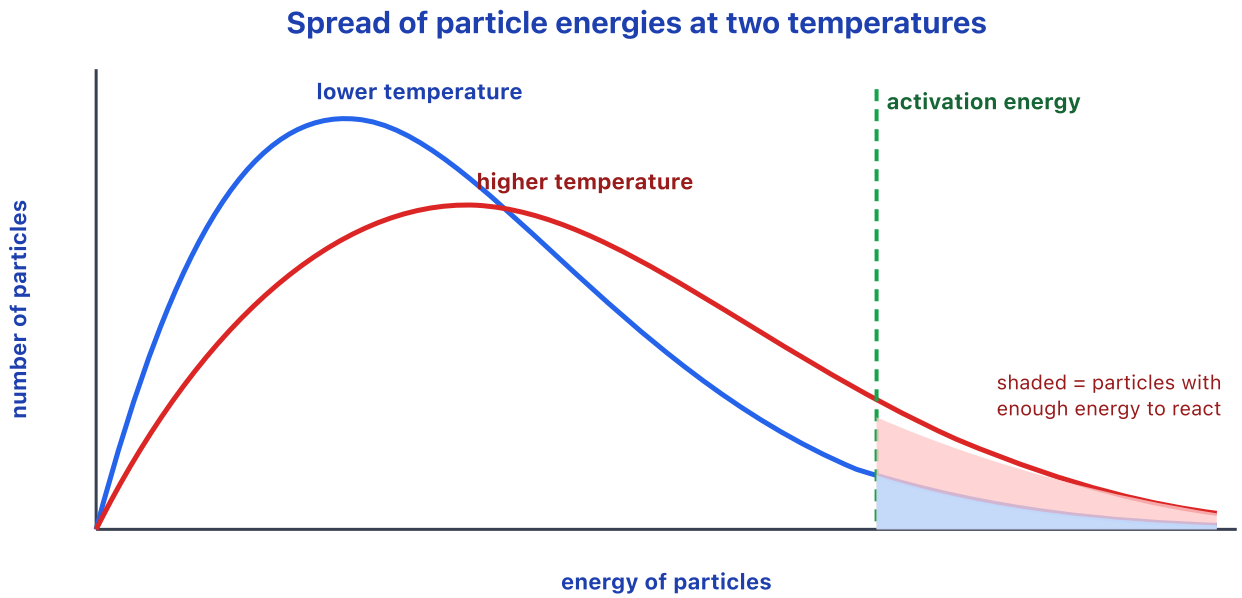
More surface exposed → more frequent collisions → faster rate.
```

Exam Tip: The most common lost mark in rate explanations is writing "more collisions" instead of "**more frequent collisions**" or "more collisions **per second**". Over the whole reaction the total number of collisions is not the point — it is how often they happen. For temperature, you must give **both** reasons: particles collide more often *and* a greater proportion of collisions have energy greater than or equal to the activation energy. The second reason is the more important one.

Temperature and the Energy Distribution of Particles

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At any temperature, particles have a spread of energies. Raising the temperature shifts the distribution so that **many more particles** have energy greater than the activation energy. This is why a rise of only about 10 °C can roughly **double** the rate of many reactions.



4.6.1.4 Catalysts

A **catalyst** is a substance that **changes the rate** of a chemical reaction (usually increases it) but is **not used up** during the reaction. Different reactions need different catalysts.

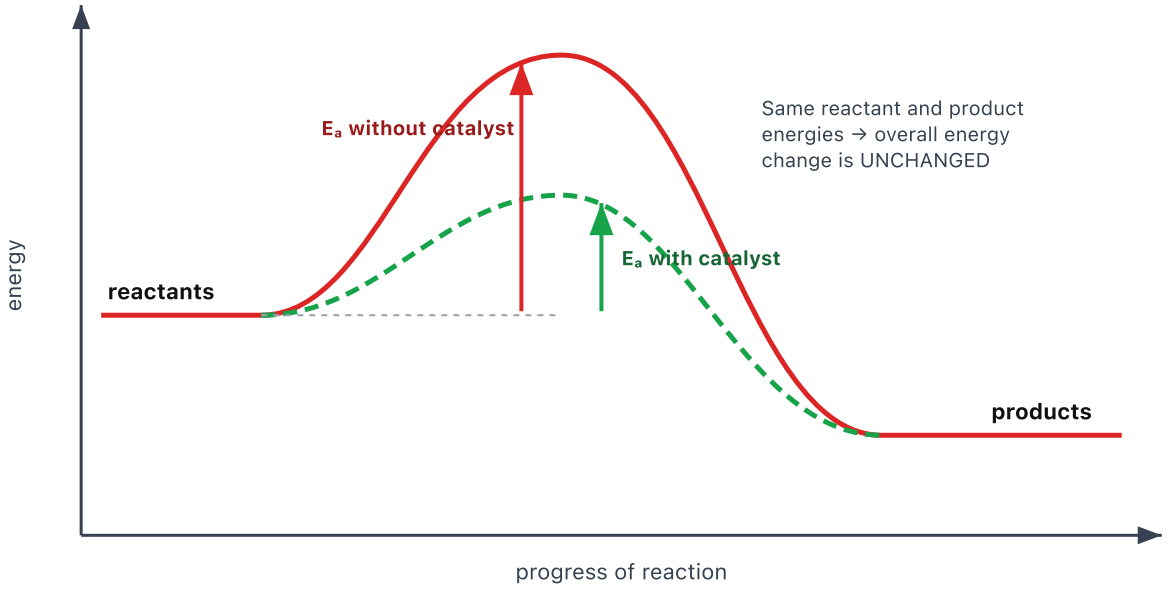
Catalysts increase the rate of reaction by providing a **different pathway** for the reaction that has a **lower activation energy**.

Enzymes act as catalysts in biological systems.

Because the catalyst is not used up, it is **not included in the chemical equation** for the reaction (though it may be written above the arrow). The same mass of catalyst remains at the end and can be reused. Only a small amount is needed.

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Reaction profile with and without a catalyst



Catalyst	Reaction catalysed
Manganese(IV) oxide, MnO_2	Decomposition of hydrogen peroxide: $2H_2O_2 \rightarrow 2H_2O + O_2$
Iron	Haber process: $N_2 + 3H_2 \rightleftharpoons 2NH_3$
Vanadium(V) oxide, V_2O_5	Contact process (making sulfuric acid): $2SO_2 + O_2 \rightleftharpoons 2SO_3$
Platinum / rhodium / palladium	Catalytic converters in car exhausts
Copper(II) sulfate	Speeds up zinc + sulfuric acid
Enzymes (e.g. amylase, catalase)	Digestion of starch; breakdown of hydrogen peroxide in cells; fermentation (yeast enzymes)

You should be able to **identify catalysts** in reactions from their effect on the rate of reaction and because they are **not included in the chemical equation** for the reaction. For example, if adding a little copper(II) sulfate makes zinc react much faster with acid and the copper(II) sulfate is still present at the end, it is acting as a catalyst.

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Industrial catalysts are valuable because they allow reactions to proceed quickly at **lower temperatures**, which saves energy and money and reduces CO_2 emissions from burning fuels. They are often used as fine powders or meshes to maximise their **surface area**.

> **Exam Tip:** "Explain how a catalyst increases the rate" — you must say it **provides an alternative reaction pathway with a lower activation energy**. Saying "it lowers the activation energy" alone may be accepted, but saying "it gives particles more energy" or "it increases the number of collisions" is wrong. A catalyst does not change the energy of the particles; it lowers the energy barrier so that more of the existing collisions are successful.

4.6.2 Reversible Reactions and Dynamic Equilibrium

4.6.2.1 Reversible Reactions

In some chemical reactions, the products of the reaction can react to produce the original reactants. Such reactions are called **reversible reactions** and are represented with the symbol ⇌ (two half-arrows):

$$A + B \rightleftharpoons C + D$$

The direction of reversible reactions can be changed by changing the conditions.

The classic example is the thermal decomposition of **ammonium chloride**:

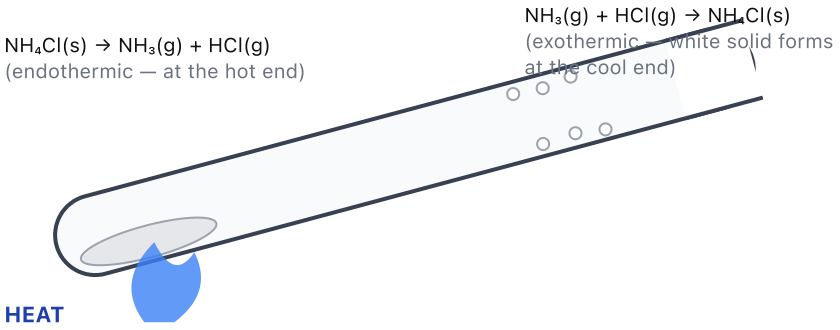
$$\text{ammonium chloride} \overset{\text{heat}}{\underset{\text{cool}}{\rightleftharpoons}} \text{ammonia} + \text{hydrogen chloride}$$

$$NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$$

On heating in a test tube, white solid NH_4Cl decomposes into two gases. These gases recombine on the cooler upper part of the tube, forming white NH_4Cl solid again.

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Heating ammonium chloride



4.6.2.2 Energy Changes and Reversible Reactions

> If a reversible reaction is **exothermic in one direction**, it is **endothermic in the opposite direction**. The **same amount of energy** is transferred in each case.

For example, the dehydration of hydrated copper(II) sulfate:

$$\underset{\text{(blue)}}{\text{hydrated copper(II) sulfate}} \xrightleftharpoons{\text{endothermic (heat)}} \underset{\text{(white)}}{\text{anhydrous copper(II) sulfate}} + \text{water}$$

$$\underset{\text{(blue)}}{CuSO_4 \cdot 5H_2O(s)} \xrightleftharpoons{\text{endothermic (heat)}} \underset{\text{(white)}}{CuSO_4(s)} + 5H_2O(l)$$

- Heating blue crystals drives off water → white powder (energy taken in).
- Adding water to the white powder turns it blue again and the mixture gets HOT (the same amount of energy is given out).
- This is used as a TEST FOR WATER: white anhydrous $CuSO_4$ turns blue.

You should be able to describe this reaction and explain why the same amount of energy is involved in each direction: the forward and backward reactions involve breaking and forming exactly the same bonds, just the other way round.

4.6.2.3 Equilibrium

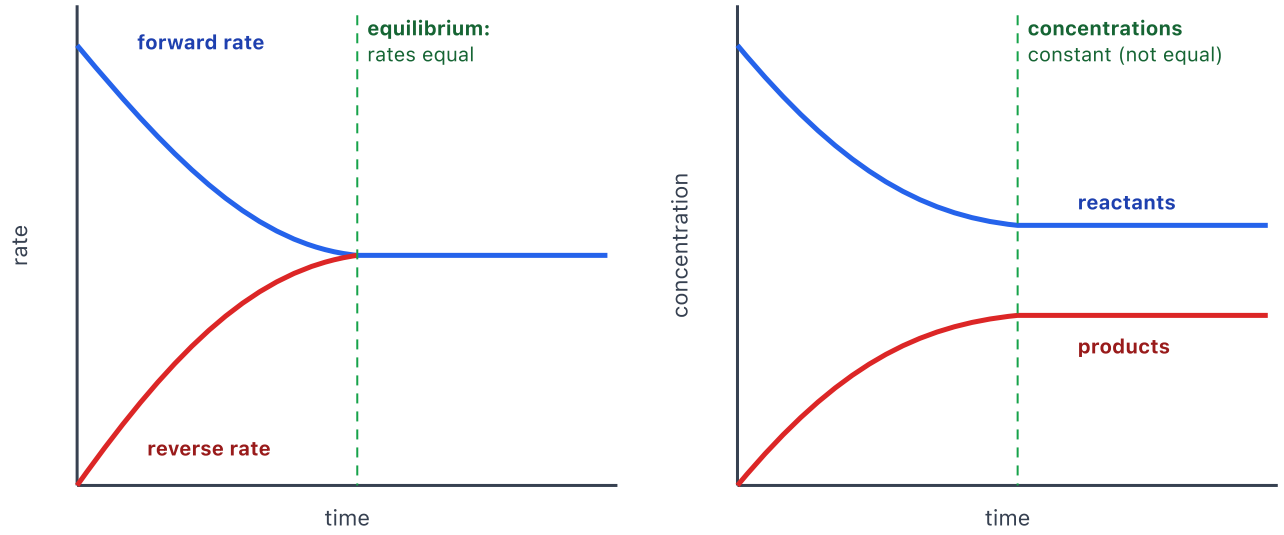
> When a reversible reaction occurs in apparatus which prevents the escape of reactants and products (a **closed system**), **equilibrium** is reached when the **forward and reverse reactions occur at exactly the same rate**.

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At equilibrium, the **concentrations** of all the reacting substances **remain constant** (but are *not* necessarily equal).

This is called **dynamic** equilibrium because both reactions are still happening — reactants are constantly being turned into products and products back into reactants — but because the rates are equal, there is no overall change.

Reaching dynamic equilibrium



Both reactions continue at equilibrium — that is why it is called dynamic.

Exam Tip: Two classic errors: (1) "at equilibrium the reaction stops" — wrong, both reactions continue at equal rates; (2) "at equilibrium the amounts of reactants and products are equal" — wrong, the **concentrations are constant**, but they can be very different from each other. Also remember that equilibrium can only be reached in a **closed system**, where nothing can enter or leave.

4.6.2.4 The Effect of Changing Conditions on Equilibrium (HT)

The relative amounts of all the reactants and products at equilibrium depend on the **conditions** of the reaction.

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> If a system is at equilibrium and a change is made to any of the conditions, then the system **responds to counteract the change**. This is **Le Chatelier's Principle**.

The effects of changing conditions on a system at equilibrium can be predicted using Le Chatelier's Principle. When the system "counteracts the change", the **position of equilibrium** shifts:

- **Shifts to the right** (forwards) → more products are formed.
- **Shifts to the left** (backwards) → more reactants are formed.

A new equilibrium is then established with different relative amounts of each substance.

4.6.2.5 The Effect of Changing Concentration (HT)

> If the **concentration of one of the reactants or products is changed**, the system is no longer at equilibrium and the concentrations of all the substances will change until equilibrium is reached again.
>
> If the concentration of a **reactant is increased**, more products will be formed until equilibrium is reached again.
> If the concentration of a **product is decreased**, more reactants will react until equilibrium is reached again.

Change	Shift in position of equilibrium	Result
Increase concentration of a reactant	Right (forwards) — uses up the added reactant	More product
Decrease concentration of a reactant	Left (backwards) — replaces the reactant	Less product
Increase concentration of a product	Left (backwards) — uses up the added product	More reactant

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Decrease concentration of a product (e.g. remove it)	Right (forwards) — replaces the product	More product

A good illustration in school uses the equilibrium between yellow chromate ions and orange dichromate ions:

$2CrO_4^{2-}(aq) + 2H^+(aq) \rightleftharpoons Cr_2O_7^{2-}(aq) + H_2O(l)$
yellow orange

Add acid (more H^+) → equilibrium shifts RIGHT → turns orange
Add alkali (removes H^+) → equilibrium shifts LEFT → turns yellow

4.6.2.6 The Effect of Temperature Changes on Equilibrium (HT)

> If the temperature of a system at equilibrium is **increased**: the relative amount of products at equilibrium **increases for an endothermic** reaction, and the relative amount of products at equilibrium **decreases for an exothermic** reaction.
>
> If the temperature of a system at equilibrium is **decreased**: the relative amount of products at equilibrium **increases for an exothermic** reaction, and the relative amount of products at equilibrium **decreases for an endothermic** reaction.

Put simply: increasing the temperature favours the **endothermic** direction (which takes in the extra energy, counteracting the change); decreasing the temperature favours the **exothermic** direction (which releases energy).

Example: $2NO_2(g) \rightleftharpoons N_2O_4(g)$ forward reaction is EXOTHERMIC
brown colourless

- Heat the sealed tube → favours endothermic (backward) direction
  → equilibrium shifts LEFT → more NO_2 → DARKER brown
- Cool in iced water → favours exothermic (forward) direction
  → equilibrium shifts RIGHT → more N_2O_4 → PALER

4.6.2.7 The Effect of Pressure Changes on Equilibrium (HT)

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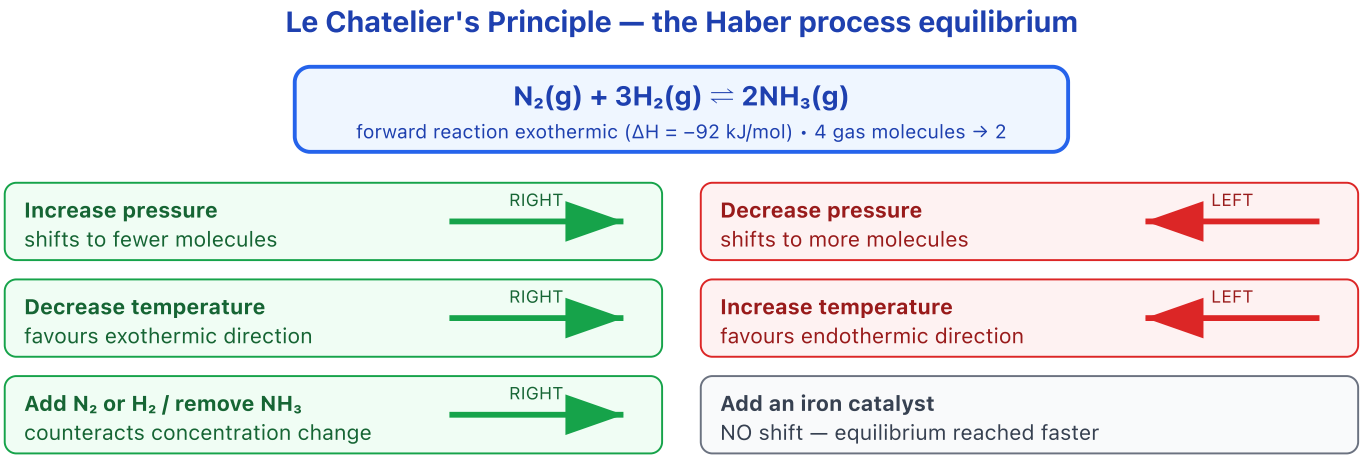
For gaseous reactions at equilibrium:
An **increase in pressure** causes the equilibrium position to shift towards the side with the **smaller number of molecules**, as shown by the symbol equation for that reaction.
A **decrease in pressure** causes the equilibrium position to shift towards the side with the **larger number of molecules**.

Count only the molecules of **gases**, using the balancing numbers in the equation. If both sides have the **same number** of gas molecules, changing the pressure has **no effect** on the position of equilibrium (though it still changes the rate).

```
N₂(g) + 3H₂(g)  ⇌  2NH₃(g)
 1    +   3     =  4 molecules on left;  2 molecules on right

Increase pressure → shifts RIGHT (fewer molecules) → MORE ammonia
Decrease pressure → shifts LEFT  (more molecules)  → LESS ammonia

H₂(g) + I₂(g)  ⇌  2HI(g)
 2 molecules   =   2 molecules  → pressure has NO effect on position
```



Green = more ammonia at equilibrium; red = less ammonia at equilibrium.

Catalysts and Equilibrium

A catalyst speeds up the forward **and** reverse reactions **equally**. It therefore has **no effect on the position of equilibrium** or on the yield — but it allows equilibrium to be reached **more quickly**. This is very valuable industrially because it saves time.

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Worked Example 4 — Predicting Equilibrium Shifts (HT)

Question: Sulfur trioxide is made in the Contact process:
$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ $\Delta H = -196$ kJ/mol
Predict and explain the effect on the yield of SO_3 of
(a) increasing the pressure, (b) increasing the temperature.

(a) Left side: 2 + 1 = 3 gas molecules. Right side: 2.
Increasing pressure shifts the equilibrium towards the side with fewer molecules → to the RIGHT → yield of SO_3 INCREASES.

(b) The forward reaction is exothermic (ΔH negative).
Increasing temperature favours the endothermic (reverse) reaction → equilibrium shifts LEFT → yield of SO_3 DECREASES.

Compromise Conditions in Industry

Conditions that give the best **yield** are not always the best **overall**. For an exothermic reaction such as the Haber process:

Condition	Effect on yield	Effect on rate	Compromise chosen
Temperature	Low temperature gives a higher yield (exothermic forward reaction)	Low temperature makes the rate too slow	About **450 °C** — a reasonable yield at an acceptable rate
Pressure	High pressure gives a higher yield (fewer gas molecules on right)	High pressure also increases rate	About **200 atm** — higher pressures are too expensive and dangerous (thicker pipes, more energy for compressors)
Catalyst	No effect on yield	Increases rate	**Iron** catalyst used

Unreacted nitrogen and hydrogen are **recycled**, and ammonia is removed by cooling it until it liquefies — removing the product also helps pull the equilibrium to the right. Full details of the Haber process are covered in 4.10 (Using Resources).

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Exam Tip: For any Le Chatelier question, use a three-part answer: (1) **identify the relevant feature** of the equation ("the forward reaction is exothermic" or "there are 4 molecules of gas on the left and 2 on the right"); (2) **state the shift** ("the position of equilibrium moves to the right"); (3) **state the effect** on the yield ("so more ammonia is produced"). Missing step (1) often loses the first mark. Never say that pressure affects reactions with no gases, or that a catalyst changes the yield.

Summary Checklist for 4.6 The Rate and Extent of Chemical Change

- Calculate mean rate from quantity of reactant used or product formed ÷ time, in g/s, cm^3/s or (HT) mol/s.
- Describe methods for measuring rate: gas syringe, mass loss, colour change/turbidity.
- Draw and interpret graphs of quantity against time; explain the shape (steep start, levelling off).
- (HT) Draw tangents and calculate their gradients to find the rate at a specific time.
- State the factors affecting rate: concentration, pressure, surface area, temperature, catalysts.
- Describe Required Practical 5: the thiosulfate "disappearing cross" (turbidity) method and the gas-volume method; identify variables.
- Use collision theory to explain each factor, using "frequency of collisions" and "energy ≥ activation energy".
- Calculate surface area to volume ratios.
- Define a catalyst; explain that it provides an alternative pathway with lower activation energy; draw a reaction profile with and without a catalyst; recall that enzymes are biological catalysts.
- Recognise reversible reactions (⇌): ammonium chloride; hydrated/anhydrous copper(II) sulfate.
- Explain that a reaction exothermic in one direction is endothermic in the other by the same amount.
- Define dynamic equilibrium: forward and reverse rates equal in a closed system; concentrations constant.
- (HT) State Le Chatelier's Principle and predict the effect of changing concentration, temperature and pressure on the position of equilibrium.

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- (HT) Explain why industrial conditions are a compromise between yield, rate, cost and safety.

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